# Treatment Effects after Endogenous Switching Regression: the `msat` Command

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**Abstract**
The Stata command `movestay` (Lokshin and Sajaia, 2004) fits the endogenous switching regression (ESR) by full-information maximum likelihood. The post-estimation command `msat` computes from its output the average treatment effects on the treated (ATT), on the untreated (ATU) and on the population (ATE), the selection-on-gains parameter $\kappa = \rho_1\sigma_1 - \rho_0\sigma_0$ with a Wald test, the expected effect by quantile of any ranking variable, and the marginal treatment effect (MTE). This note presents version 1.3. It restates the four conditional expectations and establishes the identities that organize their reading: the ATT–ATU gap is $\kappa$ times a positive factor, the Mills terms cancel in the ATE, and $\kappa$ is the slope of the MTE. It then separates what the correction needs from what the likelihood assumes: the two-step estimator (probit, then least squares of each regime on $X$ and its Mills ratio) is consistent under a normal participation error and a linear conditional mean of the regime errors, where the likelihood needs the entire joint law. On designs with skewed outcome errors the likelihood fails—correlations driven to $\pm1$, $\kappa$ tripled—and the two-step estimator does not; `msat` computes both, and their disagreement is the practical test of joint normality. A semiparametric MTE by percentile-weights regression on the propensity score, with a bandwidth chosen for a derivative, is reported next to the parametric line. Standard errors combine a design-based sampling component with a delta-method component and are validated by an independent implementation, by bootstrap with full re-estimation, and by Monte Carlo coverage on datasets with known parameters, which are distributed with the command.

**Keywords:** endogenous switching regression; treatment effects; selection on gains; marginal treatment effect; two-step estimation; percentile-weights regression; `movestay`; `msat`; Stata.

## 1 Introduction

The endogenous switching regression (ESR) of Lee (1978) and Maddala (1983) is a workhorse for program evaluation with survey data when participation is voluntary and returns are heterogeneous: two outcome equations, one per regime, are estimated jointly with a selection equation, and the correlations between the selection error and each regime's error absorb selection on unobservables. Lokshin and Sajaia (2004) implemented full-information maximum likelihood estimation of the ESR in the Stata command `movestay`. The PEP post-estimation command `msat` builds on it and is meant to be the single entry point after `movestay`: one call reports the three average effects most often needed in applied work, the parameter that separates them, and the profile of the effect along the participation unobservable.

This note presents version 1.3. Section 2 restates the model, derives the four conditional expectations and the three estimands, and establishes the identities that organize their interpretation: the ATT–ATU gap is proportional to $\kappa = \rho_1\sigma_1 - \rho_0\sigma_0$, the covariance between the selection error and the unobserved individual gain; the Mills terms cancel exactly in the ATE; and $\kappa$ is the slope of the marginal treatment effect. Section 3 contrasts the two routes to the parameters, full-information maximum likelihood and the two-step estimator, by the assumptions each needs rather than by efficiency; it is the theoretical core of the note and the reason `msat` computes both. Section 4 derives the standard errors and validates them. Section 5 presents the outputs that describe heterogeneity: $\kappa$ and its test, the effect by quantile of a ranking variable, the parametric MTE with its support flags, and the semiparametric MTE by percentile-weights regression, whose bandwidth is chosen for a derivative. Section 6 describes the command and its validation datasets, Section 6.4 its relation to Stata's `eteffects`, Section 7 the Monte Carlo illustrations, and Section 8 the recommendations for practice.

**Note on versions.** Version 1.3 supersedes version 1.0 (April 2015). The two counterfactual conditional expectations now follow equations (6) and (7) of Lokshin and Sajaia (2004), and standard errors include parameter-estimation uncertainty; estimates and standard errors from the two versions therefore differ. The statement in the 2015 note that equations (5)–(8) of Lokshin and Sajaia (2004) carry a sign error is withdrawn: those equations are correct, and the present implementation follows them.

# 2 The model and the estimands

## 2.1 Endogenous switching regression

Let $D_i \in \{0, 1\}$ be the observed regime and write the model, in the regime indexing used by `movestay`'s output (0 = untreated, 1 = treated),

$$D_i^* = Z_i'\gamma + u_i, \qquad D_i = \mathbf{1}\{D_i^* > 0\}, \tag{1}$$
$$Y_{1i} = X_i'\beta_1 + \varepsilon_{1i} \quad \text{observed if } D_i = 1, \tag{2}$$
$$Y_{0i} = X_i'\beta_0 + \varepsilon_{0i} \quad \text{observed if } D_i = 0, \tag{3}$$

with $(\varepsilon_{1i}, \varepsilon_{0i}, u_i)$ trivariate normal, $\mathrm{Var}(u_i) = 1$ (the probit normalization: only $\gamma/\sigma_u$ is identified), $\mathrm{Var}(\varepsilon_{ji}) = \sigma_j^2$, and $\mathrm{Corr}(\varepsilon_{ji}, u_i) = \rho_j$ for $j = 0, 1$. Both $\rho_0$ and $\rho_1$ are freely estimated. The covariance between $\varepsilon_{1i}$ and $\varepsilon_{0i}$ is not identified, because no unit is observed in both regimes; it does not enter the likelihood or any of the estimands below. `movestay` maximizes the likelihood over $(\beta_1, \beta_0, \gamma, \ln\sigma_1, \ln\sigma_0, \operatorname{atanh}\rho_1, \operatorname{atanh}\rho_0)$ and reports `e(b)` and `e(V)` in that parameterization.

## 2.2 The four conditional expectations

By joint normality, $\mathbb{E}[\varepsilon_{ji} \mid u_i] = \rho_j\sigma_j u_i$, so every conditional expectation reduces to $\mathbb{E}[u_i \mid D_i]$. With $\lambda_1(Z) \equiv \phi(Z'\gamma)/\Phi(Z'\gamma)$ and $\lambda_0(Z) \equiv \phi(Z'\gamma)/\{1-\Phi(Z'\gamma)\}$, the standard truncated-normal moments give $\mathbb{E}[u_i \mid D_i = 1] = \lambda_1$ and $\mathbb{E}[u_i \mid D_i = 0] = -\lambda_0$ (Appendix A). Hence

$$\begin{aligned} \mathbb{E}[Y_{1i} \mid D_i = 1] &= X_i'\beta_1 + \rho_1\sigma_1\lambda_1, & \mathbb{E}[Y_{0i} \mid D_i = 1] &= X_i'\beta_0 + \rho_0\sigma_0\lambda_1, \\ \mathbb{E}[Y_{1i} \mid D_i = 0] &= X_i'\beta_1 - \rho_1\sigma_1\lambda_0, & \mathbb{E}[Y_{0i} \mid D_i = 0] &= X_i'\beta_0 - \rho_0\sigma_0\lambda_0. \end{aligned} \tag{4}$$

These are equations (5)–(8) of Lokshin and Sajaia (2004), in the present indexing. The rule they embody is worth stating on its own, because it is the point at which implementations most easily go wrong:

> *The sign of the Mills correction is fixed by the conditioning subsample—$+\lambda_1$ on the treated, $-\lambda_0$ on the untreated—and does not depend on which regime's outcome is being predicted.* The regime index selects $\beta_j$, $\sigma_j$, $\rho_j$; the subsample selects the sign.

### 2.3 The three estimands

Averaging over the relevant subsample, with $\Delta_i = Y_{1i} - Y_{0i}$,

$$\text{ATT} = \mathbb{E}[\Delta_i \mid D_i = 1] = \mathbb{E}\big[X_i'(\beta_1 - \beta_0) + \kappa\lambda_1 \,\big|\, D_i = 1\big], \tag{5}$$

$$\text{ATU} = \mathbb{E}[\Delta_i \mid D_i = 0] = \mathbb{E}\big[X_i'(\beta_1 - \beta_0) - \kappa\lambda_0 \,\big|\, D_i = 0\big], \tag{6}$$

$$\text{ATE} = p\,\text{ATT} + (1-p)\,\text{ATU}, \qquad p = \Pr(D_i = 1), \tag{7}$$

where

$$\kappa \;\equiv\; \rho_1\sigma_1 - \rho_0\sigma_0 \;=\; \text{Cov}(\varepsilon_{1i} - \varepsilon_{0i},\, u_i) \tag{8}$$

is the covariance between the selection error and the unobserved individual gain. Two identities organize everything that follows.

**Proposition 1** (The ATT–ATU gap is selection on gains)**.** *At common $X$ and $Z$,* $\text{ATT}-\text{ATU} = \kappa\,[\lambda_1(Z) + \lambda_0(Z)]$*, with $\lambda_1 + \lambda_0 > 0$. Hence* $\text{sign}(\text{ATT} - \text{ATU}) = \text{sign}(\kappa)$*, and* ATT $=$ ATU *if and only if $\kappa = 0$.*

The endogenous treatment regression (Stata's `etregress`) specifies a single outcome equation with a common error and a single $\rho$; it therefore imposes $\kappa \equiv 0$ and cannot, by construction, produce different ATT and ATU. The ESR's ability to estimate $\kappa$ is its distinctive contribution.

**Proposition 2** (The Mills terms cancel in the ATE)**.** *Since $\Phi(Z'\gamma)\lambda_1(Z) = \phi(Z'\gamma) = \{1 - \Phi(Z'\gamma)\}\lambda_0(Z)$, one has $p\,\mathbb{E}[\lambda_1 \mid D = 1] = (1-p)\,\mathbb{E}[\lambda_0 \mid D = 0] = \mathbb{E}[\phi(Z'\gamma)]$, and therefore*

$$\text{ATE} = \mathbb{E}[X_i]'(\beta_1 - \beta_0).$$

The selection correction is thus invisible in the formula for the ATE but load-bearing in its estimation: the ATE is correct only insofar as $\hat\beta_1$ and $\hat\beta_0$ are consistent, which is what the Mills terms secure. The choice among ATT, ATU and ATE is, once the $\beta$'s are purged, a choice of the population over which $X$ is averaged.

### 2.4 The marginal treatment effect

Let $u_D = \Phi(-u)$ be the participation unobservable on the unit interval, so that $D = 1$ if and only if $u_D < P(Z) \equiv \Phi(Z'\gamma)$; small $u_D$ means eager to participate. The marginal treatment effect of Heckman and Vytlacil (2005) is the expected gain of the units at the margin, $\text{MTE}(x, u_D) = \mathbb{E}[\Delta \mid X = x, u_D]$. Under the ESR, $\mathbb{E}[\varepsilon_1 - \varepsilon_0 \mid u] = \kappa u$ and $u = \Phi^{-1}(1 - u_D)$, hence

$$\text{MTE}(x, u_D) = x'(\beta_1 - \beta_0) + \kappa\,\Phi^{-1}(1 - u_D). \tag{9}$$

The curve is a straight line in $\Phi^{-1}(1-u_D)$ with slope $\kappa$: it is flat if and only if there is no selection on gains, and the ATT, ATU and ATE are its integrals against the weights of Heckman and Vytlacil (2005). Two consequences matter for the command. The MTE at $u_D$ is identified by the data only where $P(Z)$ takes the value $u_D$ in both regimes; outside the common support of $P(Z)$ the line is an extrapolation of the normal model, and `msat` flags it as such. And since (9) is linear in $\kappa$ and in $\mathbb{E}[X]'(\beta_1 - \beta_0)$, its standard error at any $u_D$ follows from the joint delta-method variance of these two quantities.

Three sources of heterogeneity should be kept apart, because the ESR identifies two of them. The observable one, $x'(\beta_1 - \beta_0)$, varies with $X$; the unobservable one correlated with participation, $\kappa u$, is what $\kappa$ measures and what the MTE displays; the unobservable one orthogonal to participation has a variance that depends on $\text{Cov}(\varepsilon_1, \varepsilon_0)$ and is not identified, so that quantiles of the individual gain are not identified without further assumptions (Heckman et al., 1997).

## 3 Two routes to the parameters

Everything in Section 2 is a statement about conditional expectations, and it holds under assumptions weaker than the trivariate normal. Write them separately.

(A1) $u_i \sim N(0,1)$, independent of $(X_i, Z_i)$: the participation equation is a probit.

(A2) $\mathbb{E}[\varepsilon_{ji} \mid u_i, X_i, Z_i] = \rho_j \sigma_j \, u_i$ for $j = 0, 1$: the regime errors are mean-linear in the participation error.

(A3) $(\varepsilon_{1i}, \varepsilon_{0i}, u_i)$ is trivariate normal.

(A3) implies (A1) and (A2); the converse is false. The four conditional expectations (4), Propositions 1 and 2, and the MTE line (9) need only (A1) and (A2): the Mills ratios come from (A1), the coefficients $\rho_j \sigma_j$ from (A2). Nothing in the effects requires the marginal law of the regime errors to be normal, or their conditional variance to be constant.

### 3.1 Full-information maximum likelihood

`movestay` maximizes the likelihood of $(Y_i, D_i)$ under (A3), over the vector

$$\theta = (\beta_1, \beta_0, \gamma, \ln \sigma_1, \ln \sigma_0, \operatorname{atanh} \rho_1, \operatorname{atanh} \rho_0).$$

Under (A3) it is efficient. Under a violation of (A3) that preserves (A1)–(A2)—skewed or heavy-tailed regime errors with a linear conditional mean—it is inconsistent, and the inconsistency is not a small one: the likelihood exploits the tails of the regime errors to fit the correlations, and pushes $\hat{\rho}_j$ towards $\pm 1$. On the design of Section 7.3, with shifted log-normal errors of skewness 1.7, $\hat{\rho}_0 = -0.92$ and $\hat{\rho}_1 = 0.93$ for true values $-0.30$ and 0.60, $\hat{\kappa} = 2.88$ for 1.08, and the ATU is off by 1.3. The selection coefficients are distorted too (the coefficient of the instrument falls from 0.88 to 0.50), so that the propensity score, the support interval and the MTE flags computed from the likelihood are themselves wrong.

### 3.2 The two-step estimator

The two-step estimator of Heckman (1979) and Lee (1978) uses (A1)–(A2) and nothing else. A probit of $D$ on $Z$ gives $\hat{\gamma}$ and the Mills ratios $\hat{\lambda}_1$, $\hat{\lambda}_0$; least squares of $Y$ on $X$ and $\hat{\lambda}_1$ among the treated gives $\hat{\beta}_1$ and $\widehat{\rho_1 \sigma_1}$; least squares of $Y$ on $X$ and $-\hat{\lambda}_0$ among the untreated gives $\hat{\beta}_0$ and $\widehat{\rho_0 \sigma_0}$; and $\hat{\kappa}$ is the difference of the two Mills coefficients. The effects follow from (5)–(6) with these estimates.

**Proposition 3** (What the two routes need)**.** *Under (A1)–(A2), the two-step estimator of* $(\beta_1, \beta_0, \gamma, \rho_1\sigma_1, \rho_0\sigma_0)$ *is consistent, and so are the ATT, ATU, ATE and* $\kappa$ *built from it. Under (A3), full-information maximum likelihood is consistent and efficient. Under (A1)–(A2) without (A3), the likelihood estimator is in general inconsistent.*

The proof of the first claim is the observation that under (A1)–(A2) the regression of $Y$ on $X$ and the regime's Mills ratio has a conditional mean that is exactly linear in its regressors, so that least squares is consistent with a generated regressor whose first stage is a correctly specified probit. The third claim is the misspecification of the likelihood.

### 3.3 What each route loses when the conditional mean bends

If (A2) fails—the conditional mean of the gain is nonlinear in $u$—both routes are misspecified, but not in the same way. The two-step regressions are projections: least squares of $Y$ on $X$ and the Mills ratio reproduces the mean of $Y$ at the mean of the regressors in each regime whatever the true form, so the regime means, and with them the ATT, ATU and ATE, remain close to their targets; what is lost is the slope, $\hat{\kappa}$, which becomes the best linear approximation of a curve. The likelihood absorbs the nonlinearity into the variance and correlation of the regime in which it occurs and loses the counterfactual mean of that regime. On the design of Section 7.3 in which the conditional mean of the gain contains a cubic Hermite term, the two-step ATU is unbiased and its $\hat{\kappa}$ falls to 0.79 for 1.08; the likelihood ATU is off by 0.87. In that case neither line is the MTE, and only a semiparametric estimator follows the curve (Section 5.4).

### 3.4 What `msat` does with this

`msat` computes the effects from `movestay`'s estimates by default, and from the two-step estimates with the option `twostep`, which re-reads the variable lists and weights of the `movestay` call, refits the probit and the two regressions, and lays the estimates out in the same parameter vector so that every downstream computation is shared. The two tables are meant to be read together: agreement is evidence for (A3); disagreement that grows with the distance of $\hat{\rho}_j$ from the two-step values is the signature of a departure from it. Under `twostep` the propensity score, the common support and the MTE flags come from the probit.

## 4 Standard errors

### 4.1 Two components

Write the estimated effect as an average of individual predicted effects,

$$\widehat{\mathrm{ATT}} = \frac{1}{n_1} \sum_{i:D_i=1} g(X_i, Z_i; \hat{\theta}), \qquad \theta = (\beta_1, \beta_0, \gamma, \ln\sigma_1, \ln\sigma_0, \operatorname{atanh}\rho_1, \operatorname{atanh}\rho_0),$$

with $g$ the bracketed term in (5). Its variance has two sources: the sampling of the covariates over which $g$ is averaged, computed by `svy: ratio` so that any survey design is honoured, and the estimation error in $\hat{\theta}$. The second is a delta-method term,

$$\mathrm{Var}_\theta(\widehat{\mathrm{ATT}}) \approx G\, \mathtt{e(V)}\, G', \qquad G = \frac{\partial\, \widehat{\mathrm{ATT}}}{\partial \theta'}, \tag{10}$$

where $G$ is obtained by central differences on the $2p+1$ evaluations of the effect at $\hat{\theta} \pm h e_j$, in the same parameterization as `e(V)`. `msat` reports the sum of the two components and stores each separately. The same gradient gives the variance of $\hat{\kappa}$, of $\mathbb{E}[X]'(\beta_1 - \beta_0)$, of their covariance, and hence of the MTE at any percentile, and the variance of every quantile-group mean of Section 5.

Under `twostep` the parameter vector collects the two regime regressions and the probit, with the coefficients on the Mills ratios in place of the variance and correlation parameters, and `e(V)` is replaced by the block-diagonal matrix of the three estimations, each regression with a robust variance. This ignores the covariance induced by the generated regressor; the delta-method component is then an approximation, and the bootstrap helper with the `twostep` option gives the exact standard errors. On `esr_skew` ($n$ = 10,000) the block-diagonal standard errors are 0.052, 0.070, 0.046 and 0.065 for the ATT, ATU, ATE and $\kappa$, against 0.054, 0.075, 0.045 and 0.074 from a 100-replication bootstrap with full re-estimation: the approximation is 4 and 7 percent short on the ATT and ATU, on the mark for the ATE, and 12 percent short on $\kappa$, the quantity most exposed to the first-stage uncertainty. It is adequate for reading the tables; the bootstrap is the variance to report.

### 4.2 Monte Carlo coverage

Table 1 reports, for 200 replications of the ESR-compatible design of Section 7.1 with $n$ = 1000, the true standard deviation of each estimate across replications, the average delta-method standard error, its average sampling-only component, and the coverage of nominal 95% intervals.

Table 1: Standard errors and coverage, 200 replications, $n = 1000$, true effect $= 2$

| | True s.d. | Sampling component | Delta-method s.e. | Coverage |
|---|---|---|---|---|
| ATT | 0.0555 | 0.0010 | 0.0631 | 97.0% |
| ATU | 0.0548 | 0.0015 | 0.0584 | 96.5% |
| ATE | 0.0447 | 0.0012 | 0.0497 | 94.5% |

Table 2: Four estimates of the standard error, `esr_annex2` ($n = 5000$)

| | Delta, Stata | Delta, Python | Bootstrap-200 | Monte Carlo s.d. |
|---|---|---|---|---|
| ATT | 0.0292 | 0.0296 | 0.0275 | 0.028 |
| ATU | 0.0267 | 0.0257 | 0.0287 | 0.025 |
| ATE | 0.0231 | 0.0232 | 0.0219 | — |

Point estimates: 1.9812, 1.9822, 1.9815 (identical in all four).

The delta-method standard errors are slightly conservative (the variance estimator used in the independent implementation is of the outer-product type; `movestay`'s analytic Hessian narrows the gap) and their coverage is close to nominal. The sampling component alone is smaller than the true standard deviation by a factor of about fifty in this design and by a factor of about four on the heterogeneous-effect dataset `esr_sim`. It measures only the dispersion of the individual predicted effects, which vanishes as the effect becomes homogeneous; a variance estimator built on it alone is therefore least reliable precisely when the model is well specified.

### 4.3 Bootstrap validation

The individual effects are smooth functions of $\hat{\theta}$, so the bootstrap over the whole procedure—re-estimating `movestay` and recomputing the effects on each resample—is valid, and provides an independent check (unlike nearest-neighbour matching, for which the bootstrap fails; Abadie and Imbens, 2008). Table 2 compares, on the $n = 5000$ validation dataset `esr_annex2`, the delta-method standard errors from `msat` in Stata, the same quantity from an independent Python implementation of the ESR likelihood and effects, a 200-replication bootstrap in Stata with full re-estimation, and the Monte Carlo standard deviation over 200 replications of the design.

All four agree to within about 8%, the precision one expects from a bootstrap of 200 replications. The bootstrap point estimates coincide with those of `msat`, confirming that the helper program re-estimates the full model on each draw.

## 5 Describing heterogeneity

### 5.1 The parameter $\kappa$ and its test

$\hat{\kappa} = \hat{\rho}_1\hat{\sigma}_1 - \hat{\rho}_0\hat{\sigma}_0$ is a smooth function of the parameter vector, and its delta-method standard error gives a Wald test of $\kappa = 0$, which is the test of selection on gains and the test of the single-equation model (`etregress`, $\kappa \equiv 0$) against the switching regression. It is distinct from the likelihood-ratio test of independent equations that `movestay` displays, which tests $\rho_1 = \rho_0 = 0$: on a design with $\rho_1 = \rho_0 = 0.5$ and a homogeneous effect, the likelihood-ratio test rejects at any level ($\chi^2_2 = 172$) while the Wald test on $\kappa$ gives $-0.001$ with a standard error of 0.026. That is selection on levels without selection on gains, and it is the case the single-equation model was built for.

## 5.2 The effect by quantile of a ranking variable

The individual expected effects $d_i = \mathbb{E}[\Delta_i \mid D_i, X_i, Z_i]$ can be averaged within quantile groups of any exogenous variable $r$, separately for the treated, the untreated and all units. Two rankings are of particular interest. Ranked on the selection index or on the score, the treated curve is $\mathrm{ATT}(a) = x'(\beta_1 - \beta_0) + \kappa\lambda_1(a)$ and the untreated curve $\mathrm{ATU}(a) = x'(\beta_1 - \beta_0) - \kappa\lambda_0(a)$: they deform in opposite directions along $a$ if and only if $\kappa \neq 0$, the treated effect being largest among those who joined against the odds and the untreated effect most negative among those who stayed out despite everything pointing them in. This is Proposition 1 decile by decile. Ranked on an initial endowment or a welfare measure, the curve is an incidence curve of expected gains along the distribution the program targets, combining observable heterogeneity with the term in $\kappa$. The ranking variable must be exogenous or pre-treatment; ranking on the outcome or on a consequence of the treatment would select on the result.

## 5.3 The parametric MTE and its support

`msat` evaluates (9) at chosen percentiles $u_D$ with the delta-method standard error and a flag stating whether $u_D$ lies inside the common support of the estimated $P(Z)$—the interval between the larger of the two group minima and the smaller of the two group maxima—or is an extrapolation. The flag is the honest complement of a parametric curve that returns a number at every $u_D$.

## 5.4 The semiparametric MTE

Under the assumptions of Heckman and Vytlacil (2005), $\mathbb{E}[Y \mid X = x, P(Z) = p] = x'\beta_0 + p\,x'(\beta_1 - \beta_0) + K(p)$ and its derivative in $p$ is $\mathrm{MTE}(x, p)$, whatever the joint law of the errors. The local instrumental variables estimator of the MTE is therefore a local regression of $Y$ on $X$ and $P$ whose coefficient on $P$ is read as a derivative. `msat` implements it with the percentile-weights regression of Araar (2016, 2026): observations are ranked on the probit score, weighted by a Gaussian kernel in rank space centred at the target percentile $\tau$,

$$\hat{w}_i(\tau) = \frac{1}{nh\sqrt{2\pi}} \exp\Big(-\frac{(\hat{p}_i - \tau)^2}{4h^2}\Big), \qquad \hat{p}_i = \hat{F}_P(\hat{P}_i),$$

and the outcome is regressed by weighted least squares on the centred covariates, their interactions with $P$, $P$ and $P^2$; the MTE at the $\tau$-quantile $q_\tau$ of the score is $\hat{\beta}_P + 2q_\tau\hat{\beta}_{P^2}$, with the influence-function standard error of Deville (1999) that accounts for the estimation of the ranks. Ranks are uniform, so the kernel effective sample size $n_{\text{eff}} \approx nh\sqrt{2\pi}$ is the same at every $\tau$; what varies is the width of the window in $p$, which is the support statistic of this estimator and is reported next to each estimate.

**A bandwidth for a derivative.** The bandwidth rule of the percentile-weights regression is tuned to a locally constant coefficient, whose variance falls as $(nh)^{-1}$ and whose MSE-optimal bandwidth is $O(n^{-1/5})$. The coefficient on $P$ is a slope inside the window: the dispersion of the regressor within the window is itself of order $h$, so its variance falls as $(nh^3)^{-1}$ and the MSE $h^4[m''(\tau)]^2 + C/(nh^3)$ is minimized at $h^* \propto n^{-1/7}$, wider than the default by a factor of three to five at survey sample sizes. `msat` therefore chooses the bandwidth in three steps: a pilot $h_0 = c\,\sigma_p n^{-1/7}$ with $c = 1.6$, the rank-space calibration of the normal-reference constant $(16/15)^{1/7}$ for a first derivative; the curvature $m''(\tau)$ of the target by second differences of a pilot curve at bandwidth $2h_0$; and a search over $h = c'h_0$, $c' \in \{0.5, \ldots, 4\}$, of the empirical criterion $h^4[\hat{m}'']^2 + \widehat{\mathrm{Var}}(\hat{\beta}_P + 2q_\tau\hat{\beta}_{P^2}; h)$, the variance being the influence-function variance evaluated at each $h$. In Monte Carlo on the designs of Section 7, the rule divides the root mean squared error of the MTE by 2.5 to 3 relative to the default bandwidth, and matches the best fixed bandwidth without knowing it. The criterion is a pointwise one: on a profile whose curvature is weak relative to the noise of a derivative it smooths the curvature away, which is what minimizes the error at each point; the global test of linearity of $\mathbb{E}[Y \mid X, P]$ in $P$ (Heckman et al., 2010) is the instrument for deciding whether the profile is flat, the semiparametric curve the instrument for saying where it is not.

**Which percentile.** The percentile of the participation unobservable gives the MTE; the percentile of the outcome distribution gives the quantile treatment effect, a difference between two marginal distributions and not the effect for any unit; and the percentile of the gain itself is not identified, since the quantiles of $Y_1 - Y_0$ require the joint law of the two outcomes that the ESR leaves free. `msat` reports the first.

# 6 The command

## 6.1 Syntax

```
msat treatvar [if] [in] [, twostep depvar(varname) hsize(varname) expand(yes)
     generate(newvar) pscore(newvar) rank(varname) nq(#) mte(numlist) semipar
     graph gropts(string) nose step(#) level(#)]
```

`msat` follows `movestay`; *treatvar* is the switching variable of its `select()` option. `twostep` selects the two-step estimates of Section 3; `depvar()` names the outcome if it cannot be recovered from `movestay`'s equation names. `generate()` saves the individual expected effects and `pscore()` the estimated probability of treatment. `rank()` and `nq()` produce the effect by quantile of a ranking variable with standard errors; `mte()` the parametric MTE at the listed percentiles with support flags; `semipar` adds the semiparametric MTE of Section 5.4, computed by the engine `_msat_pwr` distributed with the command; `graph` draws the curves. `hsize()` weights observations (for instance by household size) in addition to any `svyset` weights. `expand(yes)` exponentiates the conditional expectations before differencing, for outcomes modelled in logs; note that this yields the conditional geometric mean, and that a retransformation to the arithmetic mean requires a smearing factor (Duan, 1983). `nose` reports point estimates only, skipping the gradient and the `svy: ratio` calls; it halves the running time and is the path used inside the bootstrap. `step()` sets the relative step of the numerical gradient (default $10^{-4}$). `msat` holds and restores `movestay`'s estimation results, so it can be called repeatedly and other post-estimation commands remain available.

The command stores `r(att)`, `r(atu)`, `r(ate)`, `r(kappa)`, `r(m)` $= \mathbb{E}[X]'(\hat{\beta}_1 - \hat{\beta}_0)$, their standard errors `r(se_*)`, the two variance components `r(vd_*)` (delta) and `r(vs_*)` (sampling), the support bounds `r(p_lo)`, `r(p_hi)`, the gradient matrix `r(G)`, and the matrices `r(curve)`, `r(mte)` and `r(mte_sp)` of the heterogeneity outputs.

## 6.2 Bootstrap helper

The program `_msat_one` mirrors `movestay`'s syntax and re-estimates the model on each resample:

```
bootstrap att=r(att) atu=r(atu) ate=r(ate) kappa=r(kappa), reps(200) seed(#): ///
    _msat_one depvar indepvars [if] [in], select(treatvar [=] varlist_s) [xi twostep
    ]
```

With `cluster()` or `strata()` among the `bootstrap` options the resampling honours the survey design. If the sample is to be trimmed to the common support of the propensity score (Section 8), the trimming should be done inside the resampled program, so that every data-dependent step is within the loop.

Table 3: `msat` and `eteffects` on `esr_annex2` ($n = 5000$, true effect $= 2$)

| | `eteffects` (control function, robust s.e.) | `msat` (FIML, delta s.e.) |
|---|---|---|
| ATET / ATT | 1.957 (0.039) | 1.981 (0.029) |
| ATE | 1.975 (0.027) | 1.982 (0.023) |

### 6.3 Validation datasets

Four datasets with known parameters are distributed with the command, together with two do-files, `msat_validate.do` and `msat13_test.do`, which re-run the checks of Sections 4 and 7.

`esr_sim.dta` ($n = 20{,}000$) is drawn from the ESR model with $\rho_1 = 0.6$, $\rho_0 = -0.3$, $\sigma_1 = 1.3$, $\sigma_0 = 1.0$ and an excluded instrument. It contains both potential outcomes, so that the true ATT (1.157), ATU ($-0.313$) and ATE (0.509) are computable directly; `msat` returns 1.179 / $-0.344$ / 0.508 with delta-method standard errors 0.035 / 0.038 / 0.028.

`esr_annex2.dta` ($n = 5000$) has three regions, an unobserved `age` confounder, and an instrument, and the model is true: `age` is normal and enters both equations linearly, the selection index is linear with a normal error, and overlap is complete. The effect is homogeneous, so ATT = ATU = ATE = 2, $\rho_1 = \rho_0$, and $\kappa = 0$: the Mills correction must vanish exactly. The Stata code that generates it is in Appendix C.

`esr_nonlin.dta` and `esr_skew.dta` ($n = 10{,}000$ each) share the selection equation of `esr_sim` and $\kappa = 1.08$, and break joint normality in one way each: in the first the conditional mean of the gain contains a cubic Hermite term, $\mathbb{E}[\varepsilon_1 - \varepsilon_0 \mid u] = \kappa[u + 0.5(u^3 - 3u)]$, orthogonal to $u$ so that $\kappa$ is unchanged in projection; in the second the regime errors are shifted log-normal with skewness about 1.7 and the gain is linear in $u$. Both carry the true individual gains. They are the material of Section 7.3.

### 6.4 Relation to `eteffects`

Stata's official `eteffects` (StataCorp, 2023) estimates treatment effects with an endogenous binary treatment by a control-function regression-adjustment approach following Wooldridge (2010). A probit is fitted for treatment; its residual $\hat{\nu}_i = t_i - \Phi(z_i'\hat{\pi})$ is then entered linearly into a separate outcome equation for each treatment level, $E(y_{ij} \mid x_i, \nu_i, t_i = j) = x_i'\beta_{1j} + \nu_i\beta_{2j}$; and the parameters, the potential-outcome means, and the ATE or ATET are estimated jointly by stacked GMM with robust standard errors. The identifying assumption is that the outcome errors are mean-linear in the treatment residual, $E(\varepsilon_{ij} \mid \nu_i) = \nu_i\beta_{2j}$.

This is related to, but distinct from, the endogenous switching regression. Under the trivariate-normal ESR the conditional mean of the outcome error is proportional to the inverse Mills ratio, $E(\varepsilon_{ji} \mid t_i, z_i) = \rho_j\sigma_j\{t_i\lambda_1 - (1 - t_i)\lambda_0\}$, which is not a linear function of $t_i - \Phi(z_i'\pi)$; the two control functions coincide only approximately. The two estimators are therefore consistent under different assumptions: `eteffects` does not require joint normality but does require linearity in the treatment residual, while `movestay` requires normality and delivers the Mills-ratio correction exactly. `eteffects` accommodates nonlinear outcome models (probit, Poisson, fractional) that `movestay` does not.

Table 3 compares the two on the validation data, which are drawn from the normal ESR. Both are within one standard error of the true value and of each other. `msat` is somewhat closer to the truth and has standard errors smaller by 17 to 35 percent. Three factors contribute, and the comparison does not separate them: the efficiency of maximum likelihood over GMM under correct specification; the fact that on these data the Mills-ratio correction is the true one and the linear control function an approximation; and `eteffects`'s sandwich standard errors against `movestay`'s model-based ones (`movestay ..., robust` would put them on the same footing). What the comparison does establish is that the two commands agree on data where the ESR

Table 4: ESR-compatible design, 200 replications, true effect = 2

| | $n = 1000$ | | $n = 5000$ | |
|---|---|---|---|---|
| | Mean | s.d. | Mean | s.d. |
| ATT | 1.996 | 0.063 | 2.003 | 0.028 |
| ATU | 2.001 | 0.062 | 1.997 | 0.025 |

Table 5: Selection on gains, 150 replications, $n = 3000$

| | ATT | ATU | ATT−ATU |
|---|---|---|---|
| True | 1.455 | 0.502 | 0.953 |
| ESR | 1.456 | 0.506 | 0.950 |
| ETR (`etregress`, $\kappa \equiv 0$) | 0.852 | 0.852 | 0 |
| IPW / PSM on observables | 2.054 | — | — |

holds, and that `msat`'s standard errors—validated by coverage in Section 4—are not smaller because they are understated.

`msat` is therefore the post-estimation command for users who want the FIML estimates of the normal ESR, the ATU reported directly, the correlations $\rho_0$ and $\rho_1$—hence $\kappa$—readable from the output, survey weights through `svyset` and `hsize()`, and effects on the level of a log-modelled outcome. `eteffects` is the official alternative when the outcome is nonlinear, when joint normality is doubted but linearity in the treatment residual is acceptable, or when a GMM estimator is preferred for computational robustness. Running both and checking agreement is good practice on field data.

# 7 Illustrations

## 7.1 An ESR-compatible design

Table 4 reports the ESR on the design of `esr_annex2`, in which the model is true and the effect is homogeneous ($\kappa = 0$).

The command is unbiased at both sample sizes. Here $\rho_1 \approx \rho_0 \approx +0.5$: the endogeneity is strong and correctly detected, and since the effect is homogeneous, $\kappa \approx 0$ and the Mills correction vanishes in the effects. The selection correction does its work in the estimation of $\beta_1$ and $\beta_0$, not in the aggregation (Proposition 2).

## 7.2 A design with selection on gains

Finally, Table 5 considers a design in which an unobserved ability $a \sim N(0, 1)$ raises both participation and the return to participation ($Y_1$ loads on $a$ with coefficient 1.5, $Y_0$ with 0.5), so that $\kappa > 0$ and ATT > ATE > ATU. Joint normality holds; $n = 3000$, 150 replications.

Only the ESR recovers both effects. The ETR cannot produce a gap by construction and lands between the two; matching on observables does not see the ability and overstates the ATT by 0.6. This is the design on which the ESR's distinctive contribution—estimating $\kappa$—is visible.

On `esr_sim` ($n$ = 20,000, $\kappa = 1.08$) the same parameter is visible below the aggregates. Table 6 reports `msat`'s expected effect by decile of the estimated probability of treatment, for the treated and the untreated, next to the average of the true individual gains in the same cells: the two curves deform in opposite directions along the score, and the true gains follow the estimated ones within two standard errors in every cell but the thinnest. The Wald test gives $\hat{\kappa} = 1.089$ with a delta-method standard error of 0.037 against 0.036 by bootstrap.

Table 6: Expected effect by decile of the estimated probability of treatment, `esr_sim`

| | Treated | | | Untreated | | |
|---|---|---|---|---|---|---|
| Decile of $\hat{P}(Z)$ | Estimate | s.e. | True | Estimate | s.e. | True |
| 1 | 2.075 | 0.090 | 1.847 | $-0.092$ | 0.035 | $-0.037$ |
| 2 | 1.659 | 0.068 | 1.547 | $-0.115$ | 0.038 | $-0.121$ |
| 3 | 1.532 | 0.056 | 1.568 | $-0.213$ | 0.041 | $-0.204$ |
| 4 | 1.401 | 0.050 | 1.369 | $-0.358$ | 0.045 | $-0.316$ |
| 5 | 1.249 | 0.046 | 1.188 | $-0.477$ | 0.049 | $-0.421$ |
| 6 | 1.207 | 0.041 | 1.187 | $-0.639$ | 0.054 | $-0.564$ |
| 7 | 1.129 | 0.039 | 1.146 | $-0.724$ | 0.061 | $-0.695$ |
| 8 | 1.044 | 0.037 | 1.013 | $-0.868$ | 0.068 | $-0.947$ |
| 9 | 1.021 | 0.035 | 0.975 | $-1.027$ | 0.086 | $-0.901$ |
| 10 | 1.019 | 0.033 | 1.040 | $-1.291$ | 0.122 | $-1.262$ |
| All | 1.179 | 0.034 | 1.157 | $-0.344$ | 0.038 | $-0.313$ |

Table 7: Two departures from joint normality, 100 replications, $n = 10{,}000$

| | Normal (control) | | Gain nonlinear in $u$ | | Skewed errors | |
|---|---|---|---|---|---|---|
| | bias | RMSE | bias | RMSE | bias | RMSE |
| *ATT* | | | | | | |
| Naive difference | $-0.209$ | 0.210 | $-0.209$ | 0.212 | $-0.208$ | 0.210 |
| IPW on observables | $-0.359$ | 0.360 | $-0.361$ | 0.362 | $-0.359$ | 0.360 |
| ESR, FIML | 0.003 | 0.051 | 0.007 | 0.060 | 1.255 | 1.256 |
| ESR, two-step | 0.003 | 0.051 | $-0.010$ | 0.055 | 0.001 | 0.047 |
| *ATU* | | | | | | |
| ESR, FIML | $-0.008$ | 0.054 | $-0.865$ | 1.035 | $-1.301$ | 1.303 |
| ESR, two-step | $-0.012$ | 0.060 | 0.001 | 0.104 | $-0.001$ | 0.069 |
| $\kappa$ *(true 1.08), mean (s.d.)* | | | | | | |
| ESR, FIML | 1.088 (0.051) | | 1.492 (0.465) | | 2.879 (0.054) | |
| ESR, two-step | 1.090 (0.053) | | 0.788 (0.093) | | 1.082 (0.064) | |

True ATT/ATU: $1.176/-0.354$ (normal), $1.033/-0.174$ (nonlinear), $1.178/-0.355$ (skewed). The naive difference and IPW are biased in every design because selection on unobservables is present in every design.

### 7.3 Two departures from joint normality

The designs `esr_nonlin` and `esr_skew` isolate the two ways in which (A3) can fail while (A1) holds. Table 7 reports, over 100 replications at $n = 10{,}000$, the bias and root mean squared error of the naive difference, of inverse-probability weighting on the observables, and of the two switching-regression routes, on these designs and on the normal control.

The table is Proposition 3 in numbers. Skewed errors leave the two-step estimator exact to the third decimal and break the likelihood; a nonlinear conditional mean leaves the two-step means intact and its $\kappa$ biased, and breaks the likelihood's ATU. Matching on observables is biased by a third of the effect in every design, because the unobservable is there in every design.

Figure 1 shows the same three designs through the MTE. On the normal design the three curves coincide. Under skewed errors the two-step line and the semiparametric curve coincide and the likelihood line is wrong everywhere, with the wrong slope. Under a nonlinear conditional mean both lines are wrong in the tails, in opposite directions, and only the semiparametric curve follows the bend—rounding it, since it is a local average over the window whose width the bottom panels report. On the single sample `esr_skew`, `msat` returns ATT 2.33, ATU $-1.63$, $\hat{\kappa} = 2.82$ by likelihood and ATT 1.26, ATU $-0.34$, $\hat{\kappa} = 1.17$ by two steps, for true values 1.15, $-0.33$, 1.08; the disagreement is the diagnostic.

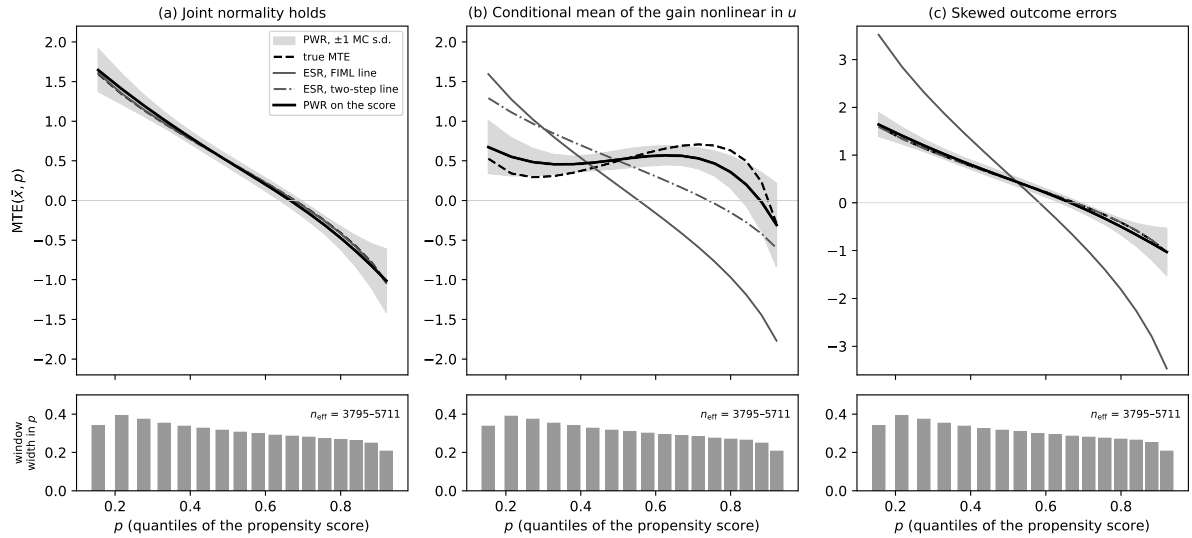


Figure 1: The MTE under three joint laws: true curve, the line of the switching regression by FIML and by two steps, and the semiparametric curve of Section 5.4 ranked on the probit score, averaged over 100 replications with ±1 Monte Carlo standard deviation. Bottom panels: width in $p$ of the kernel window at each quantile.

Table 8 is the output of `msat, twostep semipar` on the three single samples, with the bandwidth rule of Section 5.4 and Appendix B rather than the fixed bandwidth of the Monte Carlo. On `esr_sim` the semiparametric curve follows the line within one standard error except at the first decile, where it overshoots by 0.3 with a standard error of 0.19. On `esr_skew` it follows the two-step line, not the likelihood line, which is 1.7 away at the first decile and 2.2 away at the ninth; its own standard errors are two to four times those of the line, the price of not assuming the shape. On `esr_nonlin` it is flat at 0.5 over the first half of the support, where the true curve is flat, and falls in the second half, while both parametric lines cross the true curve once and are wrong by 0.5 to 1 at either end; the rule widens the window to its cap of 0.30 where the curve is flat and narrows it to 0.17 where it bends, and the bump of the true curve around $p = 0.7$ is rounded, as the Monte Carlo predicted. The kernel effective sample size stays between 3,000 and 8,000 at $n = 10{,}000$, and the width of the window in $p$ is the column to read before trusting any point: at 0.4 to 0.6 on `esr_nonlin` the estimate is an average over half the support.

# 8 Recommendations for practice

**Validate against the model's own data.** A post-estimation command should be tested on data drawn from the model it assumes, with known parameters and, where possible, known individual gains. On such data the true effects are computable directly and any discrepancy is attributable to the code; on a misspecified example, coding errors and misspecification bias can offset each other and go unnoticed. The two datasets distributed with `msat` serve this purpose.

**Run both routes.** Report the likelihood and the two-step estimates side by side. Their agreement is the evidence that joint normality can be maintained; their disagreement, especially correlations near ±1 in the likelihood output, is the evidence that it cannot, and the two-step estimates are then the ones to keep. The two-step route rests on the weaker assumption and deserves, in practice, the first place that the literature gives to the likelihood.

Table 8: `msat, twostep semipar` on the three validation samples

| $\tau$ | $p_\tau$ | True | FIML line | Two-step line | PWR (s.e.) | $h$ | $n_{\text{eff}}$ | width in $p$ |
|---|---|---|---|---|---|---|---|---|
| *`esr_sim`: joint normality holds, $n = 20{,}000$* | | | | | | | | |
| 0.10 | 0.150 | 1.62 | 1.63 | 1.62 | 1.92 (0.19) | 0.08 | 5,836 | 0.28 |
| 0.25 | 0.329 | 0.98 | 0.98 | 0.99 | 1.10 (0.09) | 0.08 | 7,691 | 0.25 |
| 0.50 | 0.582 | 0.28 | 0.28 | 0.32 | 0.30 (0.06) | 0.11 | 11,216 | 0.30 |
| 0.75 | 0.805 | −0.43 | −0.44 | −0.37 | −0.49 (0.11) | 0.11 | 10,125 | 0.27 |
| 0.90 | 0.926 | −1.06 | −1.07 | −0.97 | −0.85 (0.12) | 0.30 | 15,873 | 0.44 |
| *`esr_skew`: skewed regime errors, $n = 10{,}000$* | | | | | | | | |
| 0.10 | 0.160 | 1.57 | 3.37 | 1.71 | 2.18 (0.25) | 0.09 | 3,119 | 0.29 |
| 0.25 | 0.333 | 0.97 | 1.78 | 1.05 | 1.00 (0.10) | 0.12 | 5,412 | 0.38 |
| 0.50 | 0.580 | 0.28 | −0.01 | 0.31 | 0.36 (0.08) | 0.12 | 6,159 | 0.32 |
| 0.75 | 0.795 | −0.39 | −1.77 | −0.41 | −0.49 (0.19) | 0.09 | 4,178 | 0.20 |
| 0.90 | 0.914 | −0.98 | −3.30 | −1.05 | −0.69 (0.21) | 0.25 | 6,977 | 0.37 |
| *`esr_nonlin`: gain nonlinear in $u$, $n = 10{,}000$* | | | | | | | | |
| 0.10 | 0.160 | 0.49 | 1.55 | 1.39 | 0.57 (0.19) | 0.30 | 7,936 | 0.59 |
| 0.25 | 0.336 | 0.31 | 0.85 | 0.89 | 0.53 (0.11) | 0.17 | 6,708 | 0.57 |
| 0.50 | 0.586 | 0.61 | 0.06 | 0.33 | 0.51 (0.07) | 0.17 | 8,021 | 0.46 |
| 0.75 | 0.803 | 0.63 | −0.72 | −0.23 | 0.26 (0.16) | 0.17 | 6,709 | 0.40 |
| 0.90 | 0.921 | −0.25 | −1.41 | −0.72 | −0.14 (0.30) | 0.17 | 5,248 | 0.28 |

Stata output of `msat` 1.3.1 (do-file `msat_semipar.do`), five of the seventeen percentiles computed. $p_\tau$: the probit score at its $\tau$-quantile. True: the MTE at $p_\tau$ and at the mean of $x$, from the data generating process. FIML and two-step lines: $\hat{m} + \hat{\kappa}\,\Phi^{-1}(1 - p_\tau)$ with the estimates of each route. PWR: percentile-weights regression on the score with the derivative-aware bandwidth; standard errors by linearization, excluding the estimation of the score. $h$ in rank space, capped at 0.30.

**Include the outcome covariates in the selection equation.** Lokshin and Sajaia (2004) state that the selection equation is estimated on all exogenous variables of the outcome equations plus the instruments; the version of `movestay` used here does not add them automatically, and they must be listed in `select()`.

**Report the support of the propensity score.** The counterfactual expectations extrapolate the parametric model over the whole range of the selection index. `msat` reports the interval on which both regimes are populated and flags the MTE percentiles outside it; the ATU, which averages over the thinnest region, deserves corresponding caution. Under a failure of the likelihood the score itself is distorted, and the support should be read from the two-step output.

**Read the sign of $\kappa$.** Under voluntary adoption with comparative advantage, $\kappa > 0$ and ATT > ATE > ATU. Under targeting of the less endowed, when returns rise with endowments, $\kappa < 0$ and the ordering reverses; an estimated $\kappa > 0$ on a program designed for the poor is then evidence of capture or of misspecification. `msat` reports $\kappa$ with its standard error, and Proposition 1 gives its interpretation.

**Look at the curve, not only at the three numbers.** The effect by decile of the score, the MTE with its support flags, and the semiparametric MTE together say whether the gains are concentrated among the eager or the reluctant, whether the parametric line can be trusted at the extremes, and where the data stop and the model begins.

**Use `nose` inside the bootstrap, and bootstrap only to check.** The delta-method standard errors are the routine variance estimator; the bootstrap is a validation device and a fallback when the estimation sample is trimmed to common support inside the loop.

**In logs, smear.** `expand(yes)` returns effects on the conditional geometric mean. For effects on the arithmetic mean, multiply by the smearing factor $\frac{1}{n}\sum_i \exp(\hat{u}_i)$ (Duan, 1983); the difference is negligible when the residual variance in logs is small and material with real expenditure data.

## 9 Conclusion

`msat` 1.3 computes, after `movestay`, the ATT, ATU and ATE, the selection-on-gains parameter $\kappa$ with its test, the effect by quantile of a ranking variable and the marginal treatment effect, by full-information maximum likelihood or by the two-step estimator, with standard errors that combine a sampling component and a delta-method component and are validated by an independent implementation, by bootstrap and by Monte Carlo coverage. The theoretical point of the note is the separation of what the effects need from what the likelihood assumes: a normal participation error and a linear conditional mean of the regime errors suffice for every formula, and the two-step estimator uses only that, where the likelihood pays for its efficiency with a fragility that its output does not signal. The semiparametric MTE, with a bandwidth chosen for a derivative, is the check on both. The distinction between the situations in which each route is exact, biased or uninformative—a distinction about the errors, not about the estimator—is developed in a companion paper.

## A Derivation of the conditional expectations

For $(\varepsilon_j, u)$ bivariate normal with $\mathrm{Var}(u) = 1$, $\mathbb{E}[\varepsilon_j \mid u] = \rho_j\sigma_j u$, so $\mathbb{E}[\varepsilon_j \mid D] = \rho_j\sigma_j\,\mathbb{E}[u \mid D]$. For a standard normal $u$ and threshold $c$, $\mathbb{E}[u \mid u > c] = \phi(c)/\{1-\Phi(c)\}$ and $\mathbb{E}[u \mid u \le c] = -\phi(c)/\Phi(c)$. With $c = -Z'\gamma$ and the symmetry $\phi(-a) = \phi(a)$, $\Phi(-a) = 1-\Phi(a)$,

$$\mathbb{E}[u \mid D=1] = \frac{\phi(Z'\gamma)}{\Phi(Z'\gamma)} = \lambda_1, \qquad \mathbb{E}[u \mid D=0] = -\frac{\phi(Z'\gamma)}{1-\Phi(Z'\gamma)} = -\lambda_0,$$

which gives (4). Proposition 1 follows by subtracting (6) from (5) at common $(X, Z)$. Proposition 2 follows from $\Pr(D=1 \mid Z)\,\lambda_1(Z) = \phi(Z'\gamma) = \Pr(D=0 \mid Z)\,\lambda_0(Z)$ by taking expectations over $Z$, or directly from $\mathbb{E}[\varepsilon_1 - \varepsilon_0] = 0$ unconditionally.

## B A bandwidth for a slope in the ranking variable

Consider the percentile-weights regression of Section 5.4 with kernel $K$ in rank space, bandwidth $h$, and a regressor $P$ whose rank is the ranking variable itself. Let $m(\tau)$ be the target—the coefficient on $P$, or the derivative $\beta_P + 2q_\tau\beta_{P2}$—as a function of the percentile, and let $\hat{m}(\tau; h)$ be its estimate.

**Bias.** A kernel weight centred at $\tau$ with variance $\mu_2 h^2$ in rank space averages the target over a neighbourhood of $\tau$, and the usual expansion gives $\mathbb{E}[\hat{m}(\tau;h)] - m(\tau) = \frac{1}{2}\mu_2 h^2 m''(\tau) + o(h^2)$. For the kernel of the percentile-weights regression, $\exp\{-(p-\tau)^2/(4h^2)\}$, the variance is $2h^2$, so $\mu_2 = 2$ and the squared bias is $h^4[m''(\tau)]^2$.

**Variance of a level coefficient.** For a coefficient on a regressor whose dispersion does not depend on the window, the weighted least-squares variance is of order $\sigma^2(\tau)/(n_{\mathrm{eff}})$ with $n_{\mathrm{eff}} \propto nh$: the MSE is $h^4 m''^2 + C_0/(nh)$, minimized at $h^* \propto n^{-1/5}$. This is the rule of Araar (2026).

**Variance of a slope in the ranking variable.** The coefficient on $P$ is a slope estimated on the observations of the window, whose values of $P$ lie in an interval of width proportional to $h$ (in $p$-units, $h$ divided by the density of $P$ at $q_\tau$). The variance of a least-squares slope is the residual variance divided by the number of observations times the within-window variance of the regressor, hence of order $\sigma^2(\tau)/(nh \cdot h^2)$: the MSE is $h^4 m''^2 + C_1/(nh^3)$, and

$$h^* = \left[\frac{3C_1}{4\, m''(\tau)^2\, n}\right]^{1/7} \propto n^{-1/7}.$$

Relative to the level rule the optimal bandwidth is larger by a factor of order $n^{2/35}$ times a constant; at $n = 10{,}000$ the empirical factor is between three and five.

**Normal-reference pilot.** With a Gaussian kernel and a normal reference for the ranking variable, the plug-in constant for a first derivative is $(16/15)^{1/7} = 1.00927$, so that $h_0 = 1.00927\,\hat{\sigma} n^{-1/7}$. In rank space the ranking variable is uniform rather than normal and the target is a regression slope rather than the derivative of a function, and the constant must be recalibrated: Monte Carlo on the designs of Section 7 puts it near 1.6, which is the default of `msat`. The pilot anchors the empirical search of Section 5.4, which minimizes $h^4[\hat{m}'']^2 + \widehat{\mathrm{Var}}(\hat{m}; h)$ over multiples of $h_0$ with the influence-function variance evaluated at each $h$; the search never depends on the curvature alone, which protects it when $m'' \approx 0$, the case $\kappa = 0$.

# C The ESR-compatible design

```
set seed 20260905
set obs 5000
gen double ur = runiform()
gen byte region = 1 + (ur>0.3) + (ur>0.6)
gen double age  = 45 + 15*rnormal()            // NOT observed by the analyst
gen double educ = 1 + 4*runiform()
gen double inst = rnormal()                    // exclusion restriction
gen double idx  = -1.9 + 0.35*(region==2) + 0.45*(region==3) + 0.05*age + 0.90*inst
gen byte treatment = (idx + rnormal() > 0)
gen double income = 60 + 0.5*educ + 0.02*age + 2*treatment + 0.20*rnormal()

xi: movestay income educ , select(treatment i.region inst)   // age omitted
msat treatment
```

Expected: $\rho_0 \approx \rho_1 \approx 0.5$, ATT and ATU within 0.03 of 2, delta-method standard errors near 0.03.

**The two non-normal designs.** $x, z, u \sim N(0,1)$, $D = \mathbf{1}\{0.2 + 0.3x + 0.9z + u > 0\}$, $Y_1 = 1.0 + 1.2x + \varepsilon_1$, $Y_0 = 0.5 + 0.4x + \varepsilon_0$. Nonlinear design: $\varepsilon_0 = -0.3u + 0.95\, e_0$, $\varepsilon_1 = \varepsilon_0 + 1.08\,[u + 0.5(u^3 - 3u)] + 0.8\, e_1$, $e_j \sim N(0,1)$. Skewed design: $\varepsilon_0 = -0.3u + 0.95\, e_0$, $\varepsilon_1 = \varepsilon_0 + 1.08\, u + 0.8\, e_1$, with $e_j$ a standardized $\exp(0.5\, N(0,1))$. The normal control has $\sigma_1 = 1.3$, $\sigma_0 = 1$, $\rho_1 = 0.6$, $\rho_0 = -0.3$. In each case $\kappa = 1.08$.

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